Joaquín Fernández-León

AI Researcher · Co-founder · Senior ML Engineer

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LinkedIn · GitHub · Google Scholar · Portfolio

Profile

AI researcher and technical leader with a PhD in AI, translating research into production systems. Work spans interpretable decision trees, counterfactual explanations, insurance fraud detection, neural fields, digital twins and reinforcement learning.

Experience

Co-founder & technical AI leader — TAO Trees

May 2025 – Present

Designed TrueClaim's end-to-end ML pipeline for training, prediction, deployment and monitoring. Implemented oblique and bivariate trees; built counterfactual explanations and grounded LLM narratives for human review. Led a team of 10 and shaped product and technical roadmaps.

Computer Vision Team Lead — ARQUIMEA

Mar 2023 – May 2025

Developed computer vision and NeRF-based world models for a defence drone programme, resulting in a TRL-6 prototype. Led model development, experimentation, training and integration.

Lecturer — MSc in Data Science, UOC

Sep 2022 – Present

Teach the Deep Learning course, connecting neural network fundamentals with practical implementation.

ML Researcher & PhD Candidate — IMDEA Materials

Jun 2017 – Mar 2022

Developed deep encoder-decoder surrogate models and digital twins for composite manufacturing. Researched reinforcement learning for process control; contributed to four journal papers and an Airbus demonstrator.

Education

PhD, Computer Vision & Artificial Intelligence — Universidad Politécnica de Madrid · 2019–2023 · Cum Laude

MSc, Machine Learning & Data Science — UOC · 2018–2019

BSc, Computer Science — University of Granada · 2013–2017

MBA, Business & Digital Strategy — ThePowerMBA · 2024–2025

Selected research

An Objective, Actionable Measure of Feature Importance for Decision Trees via Counterfactual Explanations — ICAIF 2026, accepted / forthcoming.

Reinforcement learning for real-time control of Resin Transfer Molding: Bridging simulation and experiments — Composites Part A, 2026.

A digital twin for smart manufacturing of structural composites by liquid moulding — IJAMT, 2024.

A deep encoder-decoder for surrogate modelling of liquid moulding of composites — EAAI, 2023.

Technical expertise

Python, SQL, PyTorch, scikit-learn, FastAPI, Docker, Kubernetes, AWS, Databricks, MLflow, Weights & Biases, LangChain, RAG, interpretable ML, counterfactuals, neural rendering, reinforcement learning and scientific ML.